

The discharge in the vessel is supplied by means of an alternating electromagnetic field (induction) generated by an induction coil. The cylindrical coil, which forms part of the so-called power coupler, is located in a cavity inside the discharge vessel.

Power coupler. The power coupler is the heart of the induction lamp system. It transfers the energy from the HF generator to the discharge. It consists of a plastic support housing the antenna, a coaxial connecting cable and a heat conductor with mounting flange. The antenna, which is located in a cavity in the centre of the discharge vessel, includes a coil and a ferrite core, which produces a high-frequency electromagnetic field.

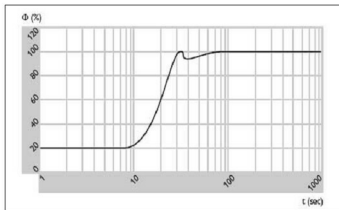


Fig. 5: Relative light output during run-up

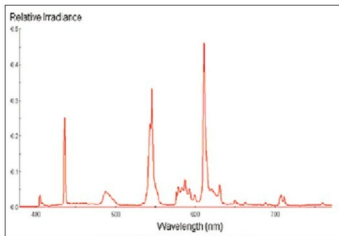


Fig. 6: The spectrum of induction lamp

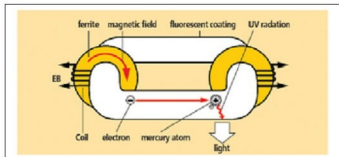


Fig. 7: Ring-shaped induction lamp. It consists of a close-loop style discharge tube with an internal fluorescent coating and one or more magnetic ferrite ring cores with coils. The electrical power for the discharge is induced from outside. The fluorescent coating converts the UV radiation emitted by mercury electrons into visible light. An electronic ballast (EB) specifically designed for the lamp is needed to operate it.

glow discharge near the wall of the power coupler cavity and this glow discharge generates sufficient number of free electrons by ionisation to release the main arc discharge, which is sustained by inductive power coupling. After ignition, the auxiliary amalgam heats up quickly and starts releasing mercury. The temperature of the bulb wall determines the mercury vapour pressure and the light output gets stabilised within 10-15 seconds.

Ignition and run up

For reliable starting in every switch, a high-voltage ignition pulse greater than 1500V for a period of approximately 100 ms, is sent through the antenna coil. The ignition pulse is powerful enough to strike the lamp. It first initiates a capacitive

Compliances

An induction lamp should meet the following standards:

1. RFI < 30MHz EN 55015
2. RFI > 30MHz EN 55022
3. Harmonics EN 61000-3-2
4. Immunity EN 61547
5. Safety EN 61347, EN 60928
6. Performance EN 60929
7. Vibration & bump tests IEC 68-2-6-Fc, IEC 68-2-29-Eb
8. Quality standard ISO 9001
9. Environmental standard ISO 14001

Operating frequencies

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Induction lamps operate at 2.6 MHz, which has been standardised in Europe. Existing in three types—separate-ballasted cavity induction lamps, external-coil induction lamps and self-ballasted cavity induction lamps—these need shielding for conducted and radiated electromagnetic interference (EMI) emissions. The reflector portion of the lamp provides enough shielding to meet FCC EMI

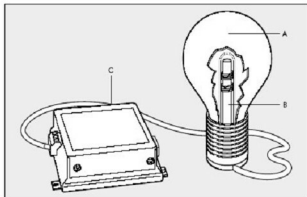


Fig. 8: A Philips QL induction lighting system, with (A) discharge vessel, (B) tube with power coupler and (C) electronic ballast